

Ravi Prakash Srivastava

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Research Interests

Large Language Models (LLMs), AI Agents, Trustworthy AI, NLP, Adversarial Machine Learning, Computer Vision.

Education

Indian Institute of Information Technology Ranchi (IIIT Ranchi), India August 2024 – May 2026
M.Tech (Data Science & AI) CGPA: 8.79/10
Institute of National Importance, Govt. of India

North Eastern Hill University (NEHU), Shillong, India August 2018 – June 2022
B.Tech (Information Technology) CGPA: 7.85/10
Central University, Govt. of India

Experience

Software Engineer *SS Software Solutions LLC, Hyderabad, India* March 2026 – Present

- Defined and enforced ML pipeline standards encompassing model versioning (MLflow/DVC), structured logging (JSON schema-validated with trace IDs for distributed tracing), and observability instrumentation (metrics, alerts, drift detection), reducing incident diagnosis time by 40% and improving pipeline reliability across 3+ production models.
- Standardized ML pipeline governance across versioning, logging, and monitoring layers, cutting untracked model deployments to zero and reducing mean-time-to-debug (MTTD) by ~35%; applied STRIDE-based Threat Modeling to identify 8+ design-level risks including data poisoning vectors and model inversion attack surfaces.
- Automated security testing by integrating SAST tools (Bandit, Semgrep) and dependency vulnerability scanning into CI/CD pipelines (GitHub Actions), reducing manual security review effort by ~60% and accelerating release cycles.

Machine Learning Intern *C-DAC (Govt. of India), Hyderabad, India* August 2022 – February 2023

- Performed large-scale data preprocessing and feature engineering on student activity logs (attendance, assignments, quizzes) to support predictive modeling.
- Implemented and benchmarked multiple machine learning models, including Decision Trees, Random Forests, and Support Vector Machines, for academic performance prediction.
- Evaluated model performance using cross-validation and metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC, identifying the best-performing model for early prediction of at-risk students.

Research Projects

- **Adversarial Prompt Injection Attacks on LLM: Cryptographic Key Leakage and Defense Strategies**
Under Publication at International Conference on Advanced Computing and Applications (ICACA-2026).
- **IoT-HITS: IoT-based Human Intrusion Detection System for Border Regions using Deep Learning**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2025.2521688>
- **Deepfake Video Detection Using Face-Centric Processing and Frame Sampling**
Under Publication International Conference on Advanced Computing and Applications (ICACA-2026).
- **Balanced Few-Shot Episodic Learning for Accurate Retinal Disease Diagnosis**
(arXiv: <https://arxiv.org/abs/2512.04967v1>)
- **Smart City Vehicle Accident Monitoring and Detection System Using MEMS, GSM, GPS Raspberry Pi 4**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2022.2043787>

Projects

Cross-Domain Text Summarization Using Transformer Models Master's Thesis

- Investigated domain shift and catastrophic forgetting by adapting **BART-large-CNN** from news summarization to legal

documents using the **BillSum** dataset.

- Implemented parameter-efficient fine-tuning (**LoRA**) with **Elastic Weight Consolidation (EWC)** to preserve pre-trained knowledge during domain adaptation.
- Achieved improved cross-domain generalization while fine-tuning only **1–2%** (~2M of 406M) of model parameters, significantly reducing GPU memory usage and training cost.

AI-Powered Vulnerability Management Platform (SAST + DAST)

Live Demo

- Developed an AI-powered vulnerability management platform integrating Static Application Security Testing (SAST) and Dynamic Application Security Testing (DAST).
- Automated static code analysis using **Semgrep** for early detection of security vulnerabilities and code quality issues.
- Implemented DAST testing for SQL Injection (SQLi), Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and other common web vulnerabilities.
- Built an interactive **Streamlit** dashboard for real-time vulnerability visualization, severity classification, and remediation recommendations.
- Designed a DevSecOps-ready architecture supporting CI/CD integration and Docker-based deployment for scalable execution.

LLM Semantic Cache & Analytics Dashboard

Live Demo

- Designed and implemented an embedding-based semantic caching layer for Large Language Models to retrieve responses using semantic similarity instead of exact keyword matching.
- Reduced LLM API latency and inference costs through intelligent cache reuse while maintaining response quality.
- Developed a real-time analytics dashboard to monitor cache hit/miss rates, similarity thresholds, query latency, and overall system performance.
- Integrated vector embeddings and similarity search to improve response retrieval efficiency for production-scale LLM applications.

Technical Skills

- **Technologies:** Python, JavaScript.
- **Databases:** SQL, MariaDB, MySQL, relational schema design.
- **Machine Learning & AI:** Deep Learning, Machine Learning, Computer Vision, Image Processing, Deepfake Detection, Large Language Models (LLMs), Hugging Face Transformers, PyTorch, Scikit-learn, OpenCV.
- **MLOps & DevOps:** Git, Docker, GitHub Actions (CI/CD), MLflow, DVC.

References

- **Dr. Ajay Kumar**
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- **Prof. Md. Iftekhhar Hussain**
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